

Samuel Boucheron Seguin

Computer Science Engineering Student · CY Tech Pau

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Seeking: a full-semester engineering internship in software or data, **mid-January to June 2027** (~5–6 months). Location open across Europe; standing interest in Scandinavia.

SUMMARY

Second-year Computer Science Engineering student at CY Tech with hands-on experience turning applied mathematics into working software — facial recognition (PCA/eigenfaces), crowd-evacuation simulation, and full-stack web apps. Served as technical lead on a 5-person Java project.

EDUCATION

CY Tech — Engineering Degree (Ingénieur), Computer Science · Pau, France 2025–2028
2nd year (ING2) · MSc in Computer Science Engineering expected 2028

Classes Préparatoires — 2-year scientific program · Pau, France 2023–2025

French Baccalauréat — Maths & Computer Science (NSI), Expert Maths option 2023
Graduated with Highest Honors (Mention Très Bien)

CS50AI: Introduction to AI with Python — HarvardX (edX) 2025–26
~100 hours · CS50 Certificate – all twelve projects, from search and logic to CNNs and NLP

SELECTED PROJECTS

Facial Recognition with PCA / Eigenfaces — Technical Lead, Java/JavaFX 2025–26 **16.41/20**

- Led architecture, code conventions, and Git/branch strategy for a 5-person team; owned integration across all modules.
- Implemented the projection and identification algorithms, including a three-criterion recognise/reject decision (Hotelling's T^2 , reconstruction error, nearest-distance) calibrated empirically on a validation set.
- Cut computation on 40,000-dimension images by computing eigenfaces from the reduced covariance matrix ($A^T A$), and evaluated accuracy with confusion matrices.

Crowd Evacuation Simulator — Top Contributor, Python 2025 **17/20**

- Built a social-force agent model (NumPy) computing pairwise repulsion forces and obstacle avoidance, animated in real time with Tkinter.
- Implemented and compared Euler vs. Runge-Kutta integration against the analytical solution to quantify numerical error.

Samedis Neige — Web App for a Sports Association, PHP 2025 **16/20**

- Lead author of the CSV processing engine (top contributor, 35/109 commits) that generated attendance sheets and ski-level groups from association data files.
- Delivered a session-secured web app (upload, processing, dated history) for the 1,213-member Amicale Laique de Billère; full lifecycle from needs analysis to demo.

heIsWatching — Browser Extension, TypeScript personal

- Built a cross-browser (Chrome/Firefox/Edge, Manifest V3) accountability extension on WXT, with a typed design-token theming system, externalized JSON content, and English/French i18n.

EXPERIENCE

Private Tutor — Toulouse & Pau 2023–2024

Tutored middle- and high-school students; built communication and client trust.

Facilities Maintenance Assistant — Cugnaux 2023

Maintained municipal facilities to hygiene/safety standards; developed autonomy.

SKILLS

LANGUAGES Java, Python, PHP, TypeScript/JavaScript, C#, C

CONCEPTS PCA/linear algebra, computer vision, numerical methods, data processing, web & frontend architecture, Git

TOOLS JavaFX, NumPy, WXT/WebExtensions, Kali Linux, UML

SPOKEN French (native), English (fluent), German (school level), Norwegian (~A1, learning)

COMPETITIONS PYR-CTF 2025 (CY Tech Pau) — first CTF; solved two challenges including a three-layer hex/Base64/Vigenère cipher